
Saturation-Like Growth Trends of SnO₂ Films for Oxide Semiconductor Applications in PEALD

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Abstract

SnO₂ plasma-enhanced atomic layer deposition (PEALD) often exhibits a gradual leveling-off of growth metrics with increasing precursor exposure time, i.e., a saturation-like behavior, which can be consistent with self-limiting growth. However, in PEALD it is difficult to rule out alternative explanations such as supply/transport limitation, plasma-induced effects, equipment drift, and spatial metrology non-uniformity. In this study, without conducting additional experiments, we use condition–result records documented in an internal SnO₂ PEALD experimental report to organize the growth behavior as a function of precursor exposure time (*source time*) using the growth per cycle (GPC) and the mean thickness as the primary metrics, and to propose an analysis framework that conservatively presents saturation-like behavior at an evidence-based level.

Under common process conditions, *source time* was varied and the GPC–*source time* relationships were compared across precursor source temperatures (40 °C and 55 °C). As a result, for both temperature conditions, GPC and mean thickness increased with increasing *source time*, while a saturation-like trend was observed in which the incremental increase was large in the initial regime and became smaller in later regimes. In addition, a condition dependence was confirmed, such that the absolute levels of the growth metrics and their increase profiles differed with precursor source temperature.

This work aims to examine whether a self-limiting growth mechanism is established and applicable in SnO₂ PEALD using TDMA-Sn as the precursor. The present analysis is expected to serve as a foundational reference for subsequent high-resolution experiments and for studies of SnO₂ oxide growth mechanisms.

1 Introduction

Oxide semiconductor thin films have become an important class of materials for display and electronic-device manufacturing, particularly in view of their compatibility with low-temperature processing, large-area uniformity, and process repeatability. Among them, SnO₂ has been widely reported as a promising oxide semiconductor owing to its wide bandgap, relatively high electron mobility, and chemical stability, enabling applications such as thin-film transistors, sensors, and transparent electrodes; nevertheless, its growth behavior and film quality can vary significantly with process conditions. [4, 8, 5] Establishing reliable process–property relationships for SnO₂ therefore requires not only controlled growth metrics but also wafer-scale thickness uniformity and reproducibility under condition changes. [2]

Atomic layer deposition (ALD) and plasma-enhanced ALD (PEALD) are widely adopted in oxide semiconductor processing because they can provide cycle-based, precise thickness control and excellent uniformity. [6] A central concept of ALD is that the surface reactions of precursors are

self-limiting. Under ideal conditions, increasing the precursor supply (e.g., exposure time) does not increase the growth beyond surface-site saturation, and the resulting saturation behavior is commonly used as an experimental indicator supporting process reliability and repeatability. [6]

However, in PEALD, plasma-generated radicals/ions/photons can act in a coupled manner during precursor adsorption and ligand removal; consequently, an observed reduction in the incremental increase of a growth metric does not necessarily imply that a self-limiting mechanism has been established. [12] For example, supply/transport limitations, plasma overreaction or surface damage, equipment-condition drift, and spatial metrology non-uniformity can all produce a “saturation-like” response. Therefore, interpreting saturation behavior in PEALD requires a conservative framework that goes beyond a qualitative judgment of “looking flat” and instead considers the extent to which observed changes are constrained relative to measurement variability (uncertainty). [2, 3]

In practical research and development settings, it is often necessary to assess what level of scientific scrutiny can be supported by accumulated experimental reports before designing additional experiments. This is particularly relevant when multiple process variables coexist and when spatial variation (within-wafer position) and equipment-condition effects may confound interpretation; in such cases, clearly delineating what can and cannot be concluded from existing records is essential for maintaining analytical credibility. [11, 13]

In this work, we analyze process conditions and measurement outcomes documented in an internal experimental report of SnO_2 PEALD, focusing on the saturation-like behavior of growth metrics with increasing precursor exposure time. The purpose of this study is not to confirm a self-limiting reaction mechanism or to claim an optimal process condition. Rather, we compare the exposure-time-dependent trends of the growth per cycle (GPC) and the mean thickness across precursor source-temperature conditions and evaluate whether the existing records provide a minimally sufficient information structure for assessing PEALD growth behavior in a conservative manner. [5]

To this end, we organize the reported condition–result entries to enable condition-wise comparison of growth behavior and record the regimes in which the incremental change of the growth metric becomes smaller as “saturation-like” regimes, thereby describing their condition dependence. This approach is intended to clarify the evidence level and interpretive boundary that can be supported by internal records while acknowledging the possibility of apparent saturation in PEALD. [12]

In addition, we used AI as a research collaboration tool to (i) structure the analysis framework, (ii) check consistency in condition–metric mapping, (iii) organize alternative explanations (potential confounders) in the interpretation, and (iv) improve logical coherence in the manuscript. All intermediate outputs generated by AI were cross-checked against the original report and reviewed by the researcher, and responsibility for the final interpretation and conclusions remains with the authors.

The remainder of this paper is organized as follows. Section 2 reviews prior work on self-limiting growth and saturation interpretation in SnO_2 ALD/PEALD. Section 3 states the objectives and scope. Section 4 details the report-based materials and methods. Section 5 reports the observed growth behavior for the compared conditions. Section 6 provides conclusions, limitations, and future directions.

2 Related Work

This section reviews how self-limiting growth and saturation behavior have been interpreted in ALD/PEALD of oxide thin films, with an emphasis on why *apparent saturation* can arise in PEALD environments. Because the present study does not aim to confirm a mechanism through new experiments, but rather to evaluate whether observations recorded in an internal experimental report provide a minimally sufficient information structure to examine *saturation-like* behavior, the discussion below focuses on *interpretive criteria and limitations* rather than a comprehensive enumeration of results. [7]

2.1 Self-limiting growth and saturation as experimental indicators in ALD

ALD is based on the assumption that precursor adsorption and surface reactions are limited by the saturation of reactive surface sites. Under ideal ALD conditions, increasing precursor exposure time (or supply) does not increase the growth per cycle (GPC) beyond a certain point, and the resulting

88 saturation response is widely used as a representative experimental indicator supporting self-limiting
89 surface chemistry. [6]

90 Nevertheless, it is generally considered conservative to avoid equating the observation of a saturation
91 curve with definitive proof of a self-limiting mechanism. In particular, the credibility of saturation-
92 based interpretation is strengthened only when measurement variability (uncertainty) and repeatability,
93 subtle process-condition changes (drift), and potential non-ideal growth contributions (e.g., parasitic
94 CVD or re-adsorption/re-deposition) are considered concurrently. [3]

95 **2.2 Growth behavior and condition dependence in SnO₂ ALD/PEALD**

96 For SnO₂ thin films, ALD/PEALD growth behavior and film properties have been reported to vary
97 substantially with the choice of precursor (e.g., TDMASn-family precursors), oxidant (thermal
98 oxidants or plasma/ozone oxidants), process temperature, and purge conditions. [1]

99 In particular, for TDMASn-based SnO₂ deposition, precursor delivery conditions (exposure time
100 and source temperature/vapor pressure) and oxidant type can simultaneously influence both GPC
101 and film characteristics. For instance, in ozone-based processes, changes in ozone concentration or
102 delivery can affect growth behavior and film properties, [10] while plasma-based processes enable
103 low-temperature deposition but introduce additional pathways by which plasma conditions may alter
104 film properties. [8]

105 Moreover, many reports emphasize that, in SnO₂ processes, film-quality metrics such as refractive
106 index, density, and electrical properties can change with process conditions even when a saturation-
107 like trend is observed in growth metrics; accordingly, saturation in growth rate alone is often regarded
108 as insufficient to claim establishment of self-limiting growth. [8]

109 **2.3 Interpretive complexity in PEALD: the possibility of apparent saturation**

110 PEALD offers enhanced reactivity at low temperature through plasma-generated active species,
111 but radicals/ions/photons can also modify surface states or alter ligand-removal pathways, thereby
112 increasing mechanistic complexity. [7]

113 As a result, even if a growth metric exhibits a reduced incremental increase at longer exposures, it is
114 difficult to isolate whether the cause is self-limiting surface saturation or alternative factors such as (i)
115 supply/transport limitation, (ii) plasma overreaction and/or surface damage, (iii) equipment-condition
116 changes and drift, or (iv) spatial metrology non-uniformity across the wafer. Such interpretive
117 uncertainty has been repeatedly discussed in broader assessments of opportunities and challenges in
118 plasma-assisted ALD. [12]

119 For SnO₂ PEALD in particular, studies incorporating in-system or post-process analyses (e.g., XPS)
120 and more detailed property characterization have been reported to interpret how process-condition
121 changes affect film states, which further supports the conservative view that a saturation-like shape
122 should not be directly attributed to self-limiting chemistry without additional corroboration. [9]

123 **2.4 The need for saturation assessment that accounts for variability**

124 Although many discussions of saturation emphasize mean growth metrics or trend shapes, practical
125 observations inevitably include variability due to within-wafer position, metrology conditions, and
126 process repeatability. In regimes where incremental growth changes become small, the magnitude
127 of the “true effect” (condition-induced change) can become comparable to measurement variability,
128 making purely qualitative judgments prone to misinterpretation.

129 Accordingly, prior work has proposed and/or employed approaches such as (i) reporting uniformity
130 and standard deviation based on wafer maps, (ii) using in-situ or in-line metrology for window
131 identification, and (iii) jointly checking for non-ideal growth contributions (e.g., parasitic CVD).
132 [2, 3] In addition, recent studies from the perspectives of process optimization and modeling explicitly
133 discuss that measurement noise (uncertainty) can influence convergence behavior and decisions
134 regarding saturation or process windows. [11, 13]

2.5 Positioning of this work

In summary, saturation behavior has been widely used as a key clue in discussing self-limiting growth in SnO₂ ALD/PEALD, and the likelihood of apparent saturation as well as the potential co-variation of film-quality indicators in PEALD have been repeatedly highlighted. [8] Recent literature also indicates growing activity in data-driven and modeling-oriented approaches, including work that addresses saturation-time prediction and the effect of noise on decision-making. [11] Therefore, it would be inappropriate to claim that the general direction of “uncertainty-aware interpretation of saturation” is entirely absent.

However, under constraints that match those of the present study (no new experiments and reliance on an internal experimental report), examples that (i) use only the limited items recorded in a legacy report, (ii) avoid arbitrary imputation or correction of missing values, and (iii) articulate a conservative, evidence-level presentation of saturation-like behavior *explicitly as a manuscript-level analysis framework* appear less foregrounded in the mainstream SnO₂ PEALD literature, which is often oriented toward process development, property optimization, or mechanism identification. [5]

Accordingly, the “gap” addressed here is not the absence of saturation interpretation per se, but the relative lack of a paper-structured, minimal-information, conservative analysis framing tailored to legacy report data. In this context, the present work aims to summarize, without over-claiming, how GPC changes with precursor exposure time within the observed range and to separate what requires subsequent verification (additional experiments, in-situ diagnostics, surface/chemical analyses) at the level of testable hypotheses. [12]

3 Study Objectives and Scope

3.1 Objectives

This study organizes SnO₂ PEALD growth behavior as a function of precursor exposure time (*source time*) using the growth per cycle (GPC) and mean thickness reported in an internal experimental record. The goal is to describe, at an evidence-based level, the observed *saturation-like* tendency in which incremental changes diminish in later exposure regimes.

Rather than claiming mechanistic confirmation or process optimality, we present condition-dependent observations across precursor source temperatures (40 °C and 55 °C) and assess whether the recorded information is sufficient to support conservative interpretation of PEALD growth trends.

3.2 Research Questions

This study addresses the following research questions:

- **RQ1.** As *source time* increases, do GPC and mean thickness increase while exhibiting a *saturation-like* tendency in which the incremental increase becomes smaller beyond a certain regime?
- **RQ2.** Do the absolute levels and the change profiles of the GPC–*source time* relationship differ with precursor source temperature (40 °C vs. 55 °C)?
- **RQ3.** (Supplementary/extension) How do auxiliary indicators included in the report (e.g., thickness standard deviation, uniformity, refractive-index range) vary alongside growth metrics, and what interpretive limitations should be imposed when using them?

3.3 Scope and Interpretive Boundaries

The scope and boundaries of interpretation are defined as follows:

1. **Data scope.** No additional experiments are conducted. The analysis is limited to process conditions and measurement outcomes recorded in the internal experimental report.
2. **Metric scope.** The primary growth metrics are GPC (Å/cycle) and mean thickness (thickness_{mean}). Thickness standard deviation, uniformity, and refractive-index range are used only as auxiliary indicators.

- 181 3. **Interpretive scope.** A saturation-like tendency is not a sufficient condition to confirm a
 182 self-limiting growth mechanism. Observed trends are presented as qualitatively consistent
 183 with self-limiting behavior, while alternative explanations—including supply/transport
 184 limitations, plasma-induced effects, and equipment drift—are not excluded.
- 185 4. **Exclusions.** Process optimization, device-level performance comparison, and identification
 186 of microscopic reaction pathways are outside the scope of this study.

187 4 Report-Based Materials and Methods

188 4.1 Nature of the Source Material and Use Principles

189 The analysis is based on an internal experimental report documenting SnO_x PEALD process set-
 190 tings and ex-situ metrology outputs. We construct a condition–outcome table to enable consistent
 191 comparisons across runs, focusing on variables relevant to Section 5.

192 Reported values are used as-is. Missing entries are retained as missing, and repeated measurements
 193 under identical or similar conditions are kept to provide limited evidence on repeatability within the
 194 available record.

195 4.2 Process Conditions and Sample Preparation

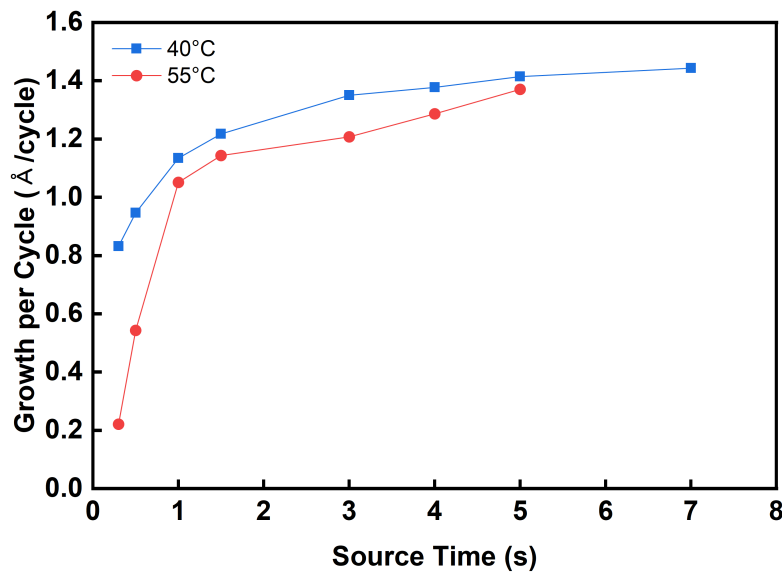


Figure 1: GPC vs. source time at two source temperatures (40°C, 55°C) under the baseline PEALD conditions.

196 According to the report, 6-inch p-type Si(100) wafers were used as substrates. Prior to deposition,
 197 BOE treatment was performed, followed by DI-water rinse and drying. SnO_2 films were deposited by
 198 PEALD using TDMA-Sn as the Sn precursor and O_2 plasma as the oxidant, with Ar purge between
 199 reaction steps. The substrate temperature was maintained at 200 °C.

200 The primary comparisons in this study follow the common baseline conditions documented in the
 201 report: chamber pressure of 2 Torr, Ar/ O_2 flow rates of 100/150 sccm, RF power of 200 W, and a
 202 total of 200 cycles.

203 4.3 Process Variables Used for Comparison

204 Section 5 focuses on the dependence of growth behavior on precursor exposure time and precursor
 205 source temperature. Accordingly, the key variables used for comparison are:

- `source_time`: TDMA-Sn exposure time (s)
- `source_temp`: TDMA-Sn source temperature (e.g., 40 °C, 55 °C)
- Fixed baseline (as documented): substrate temperature 200 °C, pressure 2 Torr, Ar/O₂ 100/150 sccm, RF 200 W, total cycles $N = 200$

Variables related to purge and plasma timing may appear in the report; however, they are not used as primary comparison axes in the main results section, and are treated as optional factors for supplementary analysis or future extensions.

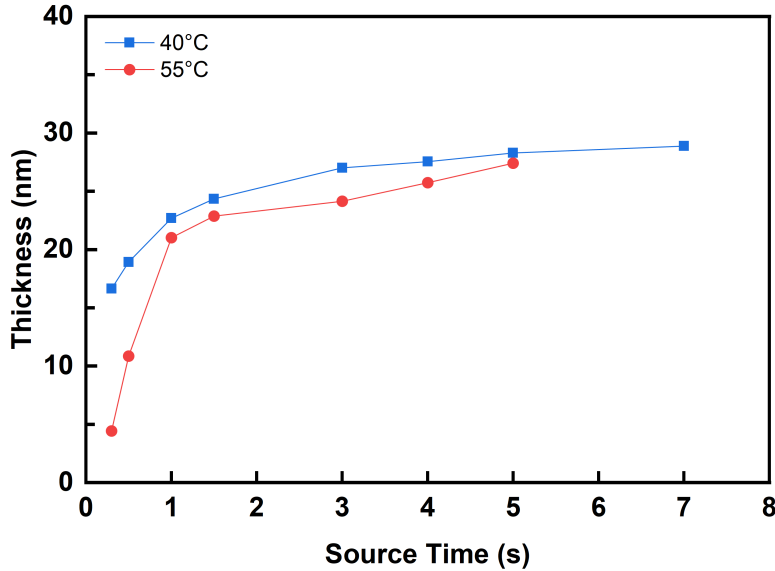


Figure 2: Mean thickness vs. source time at two source temperatures (40°C, 55°C) for $N=200$ cycles; thickness scales with GPC.

4.4 Definition of Growth Metrics and Auxiliary Indicators

Primary growth metric: GPC. The primary growth metric is the growth per cycle (GPC; Å/cycle). In the internal report, this quantity is recorded as `Dep.rate` (Å/cy), and we refer to it consistently as **GPC** in this manuscript.

When mean thickness is reported, GPC is consistent with thickness normalized by the total number of cycles:

$$\text{GPC} \equiv \frac{\text{thickness_mean}}{N}, \quad (1)$$

where $N = 200$ for the primary comparison set documented in the report.

Auxiliary indicators. Mean thickness (`thickness_mean`; Å or nm), thickness standard deviation (`thickness_std`), uniformity (%), and refractive-index range (`refractive_indexmin/max`) are treated as auxiliary indicators to contextualize potential changes in film state. Importantly, these indicators are not used to attribute changes to specific physical properties (e.g., composition, density, defect states) within the scope of this study.

4.5 Analysis Procedure

The analysis is organized around the two key plots presented in Section 5:

1. **GPC vs. source time (Figure 5-1).** For each `source_temp` condition (40 °C and 55 °C), we visualize GPC as a function of `source_time` to compare (i) the initial growth increase regime and (ii) the later regime in which the incremental increase becomes smaller (saturation-like tendency).

2. **Mean thickness vs. source time (Figure 5-2).** Under the same baseline conditions, mean thickness is plotted against `source_time`. Because the number of cycles is fixed ($N = 200$), mean thickness can be interpreted as a scaled representation of GPC ($\text{thickness} \approx \text{GPC} \times N$). This plot is used as an auxiliary consistency check to confirm that the trend observed in GPC is reflected in thickness.
3. **Logic of temperature-wise comparison.** In both plots, 40 °C and 55 °C conditions are compared on the same axes to summarize the observed differences in absolute level and trend shape. No causal attribution to a specific mechanism is made, given the absence of additional experiments or diagnostics.

4.6 AI Use and Researcher Verification

AI was used as a research collaboration tool to (i) structure the GPC-centered analysis framework, (ii) improve the logical organization of the manuscript, and (iii) set conservative interpretive boundaries so as not to over-claim self-limiting behavior from report-based, ex-situ observations. All intermediate AI-generated outputs were cross-checked against the original internal report, and the researcher retains responsibility for the final interpretation and conclusions.

5 Results and Discussion

5.1 Growth behavior as a function of source time and saturation-like tendency

Figure 1 compares the dependence of GPC (growth per cycle) on TDMA-Sn source time for two precursor source temperatures (40 °C and 55 °C). For both temperatures, GPC increases with increasing source time. The increase is more pronounced in the early regime (approximately 0.3–1.5 s in the reported condition set), whereas the incremental increase becomes progressively smaller at longer source times. In the mid-to-late regime (approximately beyond 3–5 s), the trend presents as a gradual increase with a reduced slope, consistent with a saturation-like tendency.

In this work, the observed saturation-like tendency is not interpreted as definitive evidence for the establishment of a self-limiting surface reaction. Instead, if quantitative support is required, an uncertainty-aware assessment can be performed using the criterion defined in Section 4.5 (e.g., $\Delta\text{GPC} \leq k \sigma_{\text{GPC}}$ with $k = 1$), where ΔGPC is the change between adjacent source-time conditions and σ_{GPC} is derived from within-wafer thickness variability normalized by the cycle count. Under this framework, the qualitative reduction in ΔGPC in the late regime can be reported as an evidence-level observation, while the formal classification of a saturation-like regime is reserved for the tabulated or supplementary results incorporating σ_{GPC} .

5.2 Comparison across source temperature conditions

As shown in Figure 1, differences are observed between the two source-temperature conditions. In particular, there exists a regime in which the GPC values at 40 °C are higher than those at 55 °C for the same source time. The 55 °C condition also exhibits a comparatively lower GPC at the shortest source times, followed by a steeper increase as source time increases.

While such differences may plausibly reflect combined effects related to precursor delivery (e.g., vapor pressure and flux), early-stage adsorption dynamics, and purge-related factors, the present study is limited to ex-situ observations recorded in the internal report and does not include additional diagnostics or controlled re-design of experiments. Therefore, these differences are presented strictly as condition-dependent observations rather than being attributed to a specific reaction mechanism. Where appropriate, the same conservative assessment framework (Section 4.5) may be applied to each temperature condition to compare whether the emergence and extent of saturation-like regimes differ by source temperature.

5.3 Consistency with thickness trends

Figure 2 shows the mean thickness as a function of source time under the same baseline conditions. Mean thickness increases with source time and similarly exhibits a reduced incremental increase at longer source times, consistent with the trend observed in GPC. Because the cycle count is

fixed at $N = 200$, thickness can be interpreted as a scaled representation of GPC (approximately thickness $\approx \text{GPC} \times 200$). Accordingly, Figure 2 is used as an auxiliary consistency check confirming that the GPC-based trend is reflected in the overall thickness outcome, rather than as independent evidence for a particular growth mechanism.

6 Conclusions, Limitations, and Future Work

6.1 Conclusions

This study summarizes SnO_x PEALD growth behavior as a function of precursor exposure time (source time) and precursor source temperature (source temperature) based solely on an internal experimental report (PPT). Without conducting new experiments, we use only the reported process conditions and ex-situ measurement outcomes to describe the source-time-dependent trends in GPC (growth per cycle; Å/cycle) and mean thickness, and to define a conservative interpretive boundary for reporting saturation-like behavior.

Across both source-temperature conditions (40 °C and 55 °C), GPC increases with increasing source time, with a larger incremental increase at short source times and a progressively smaller incremental increase at longer source times, forming a saturation-like tendency. Under the same fixed cycle count ($N = 200$), mean thickness exhibits a consistent trend shape, supporting a coherent comparison with the GPC-based observations. Differences in absolute GPC level and early-regime sensitivity are observed between source-temperature conditions, indicating condition dependence of growth behavior within the reported operating space.

Importantly, the saturation-like tendency observed in this report-based dataset is qualitatively consistent with growth behavior expected under self-limiting ALD/PEALD operation, and thus provides a reasonable basis for formulating the presence of a self-limiting growth regime as a central working hypothesis for process development. However, because the present study is restricted to ex-situ outcomes documented in a legacy report, the observations are not used to confirm the establishment of a self-limiting mechanism; rather, they are presented as evidence-level trends that motivate targeted verification in subsequent work.

6.2 Limitations

This study has the following limitations due to its reliance on the internal experimental report:

1. **Limited ability to confirm mechanisms.** Although the observed GPC trend is compatible with a self-limiting interpretation, alternative explanations such as precursor supply/transport limitation, plasma-induced effects, and equipment drift cannot be excluded. Accordingly, self-limiting behavior is treated as a hypothesis to be tested rather than a confirmed mechanism.
2. **Limited quantitative incorporation of variability.** The main presentation emphasizes trend comparison based on the two key plots. While within-wafer variability and repeatability information can, in principle, be used to support uncertainty-aware classification (Section 4.5), these are not visualized as error bars in the main figures; hence, any boundary of a saturation-like regime should be interpreted conservatively.
3. **Limited separation of interacting variables.** Variables beyond source time (e.g., purge- and plasma-related timings) may vary across conditions in the legacy record, limiting the ability to isolate single-variable effects. Temperature-dependent differences are therefore reported as observations without causal attribution.

6.3 Future work and process-development directions

The present results support formulating “saturation-like GPC behavior with increasing source time” as a central working hypothesis for process development. To verify and extend this hypothesis, the following steps are recommended:

1. **Strengthen repeatability assessment (Priority 1).** Perform repeated runs under identical conditions or systematically consolidate existing re-measurement records to quantify the

variability of GPC and to verify whether the late-regime reduced incremental increase persists across repeats.

2. **Define a process window via single-variable sweeps (Priority 2).** Conduct controlled sweeps in which only source time is varied while other variables are held fixed, to more clearly identify the onset and extent of the saturation-like regime. Apply the same framework across source-temperature conditions to compare the temperature dependence of the saturation-like window.

3. **Add supporting diagnostics for mechanism verification (Optional).** Where feasible, combine in-situ diagnostics (e.g., QCM or OES) or surface/chemical analyses (e.g., XPS or SIMS) to test whether the saturation-like trend corresponds to surface reaction saturation consistent with a self-limiting mechanism.

Overall, this work demonstrates that an internal experimental report can support a structured, conservative description of saturation-like growth behavior in SnO₂ PEALD at the level of observable evidence, and it provides a defensible basis for framing subsequent verification and process-development studies.

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375 AI Co-Scientist Challenge Korea Paper Checklist

376 1. Claims

377 Question: Do the main claims made in the abstract and introduction accurately reflect the
378 paper's contributions and scope?

379 Answer: [Yes]

380 Justification: The contribution and scope of interpretation are clearly defined as involving
381 the organisation of observations based on internal reports and the analysis of saturation-like
382 trends, thereby aligning with the actual content of the main text.

383 2. Limitations

384 Question: Does the paper discuss the limitations of the work performed by the authors?

385 Answer: [Yes]

386 Justification: The Discussion explicitly outlined interpretative limitations, including the
387 absence of new experiments, constraints on statistical significance, limitations in isolating
388 variable interactions, and the inability to confirm self-limiting mechanisms.

389 3. Theory Assumptions and Proofs

390 Question: For each theoretical result, does the paper provide the full set of assumptions and
391 a complete (and correct) proof?

392 Answer: [N/A]

393 Justification: This study does not present theoretical theorems/proofs, but focuses on the
394 organisation and proposed interpretative framework for the internal report's observational
395 data.

396 4. Experimental Result Reproducibility

397 Question: Does the paper fully disclose all the information needed to reproduce the main ex-
398 perimental results of the paper to the extent that it affects the main claims and/or conclusions
399 of the paper (regardless of whether the code and data are provided or not)?

400 Answer: [No]

401 Justification: Justification: Key trends were described in a reproducible format using
402 common process conditions and defined metrics (GPC, thickness, etc.). However, as the
403 source data is restricted to internal reports, external researchers face limitations in directly
404 accessing and verifying the same data.

405 5. Open access to data and code

406 Question: Does the paper provide open access to the data and code, with sufficient instruc-
407 tions to faithfully reproduce the main experimental results, as described in supplemental
408 material?

409 Answer: [No]

410 Justification: The data is based on internal experimental reports and is not provided in a
411 public repository. The main text presents the analytical procedures and metric definitions,
412 but does not provide public access to the raw data.

413 6. Experimental Setting/Details

414 Question: Does the paper specify all the training and test details (e.g., data splits, hyper-
415 parameters, how they were chosen, type of optimizer, etc.) necessary to understand the
416 results?

417 Answer: [Yes]

418 Justification: As this study involves process experiment data analysis rather than machine
419 learning training experiments, there are no items corresponding to 'training/test'. Instead, the
420 process conditions (temperature/pressure/flow rate/RF/cycles), comparison variables (source
421 time, source temperature), and metric definitions (GPC, thickness) are fully described in the
422 Methods section to provide the necessary details for interpreting the results.

423 7. Experiment Statistical Significance

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: The results were not presented in diagrammatic form, nor were statistical significance tests performed.

8. Experiments Compute Resources

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [N/A]

Justification: The core of this study lies in process experiments (PEALD) and report-based analysis; no GPU/CPU learning or large-scale computational experiments were conducted. Therefore, the reporting item "computing resources" does not directly apply to this research.

9. Code Of Ethics

Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics <https://nips.cc/public/EthicsGuidelines>?

Answer: [Yes]

Justification: This work analyzes non-human-subject PEALD process results documented in an internal report and does not involve sensitive personal data, crowdsourcing, or deployments affecting individuals. AI assistance is disclosed with researcher verification and responsibility stated, and no unsafe or deceptive use is intended.

10. Broader Impacts

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [N/A]

Justification: This study constitutes foundational research conservatively summarising growth behaviour under specific process conditions; as it is not directly linked to social deployment/decision-making systems, discussion of its societal impact is limited.

11. Safeguards

Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pretrained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: High-risk models/scraping data/deployment assets are not disclosed, and this research does not constitute a form of study where the risk of misuse within that category arises.

12. Licenses for existing assets

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [N/A]

Justification: No externally published code/datasets/models were redistributed or utilised, with the primary basis comprising internal experimental reports and literature citations. Consequently, there is no use of external assets requiring separate licensing notation.

13. New Assets

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [N/A]

Justification: This study does not distribute new datasets/code/models as public assets.

14. Crowdsourcing and Research with Human Subjects

474 Question: For crowdsourcing experiments and research with human subjects, does the paper
475 include the full text of instructions given to participants and screenshots, if applicable, as
476 well as details about compensation (if any)?

477 Answer: [N/A]

478 Justification: No crowdsourcing or human subjects research was conducted.

479 **15. Institutional Review Board (IRB) Approvals or Equivalent for Research with Human**
480 **Subjects**

481 Question: Does the paper describe potential risks incurred by study participants, whether
482 such risks were disclosed to the subjects, and whether Institutional Review Board (IRB)
483 approvals (or an equivalent approval/review based on the requirements of your country or
484 institution) were obtained?

485 Answer: [N/A]

486 Justification: As this is not human subjects research, IRB approval/notification/risk discus-
487 sion does not apply.